

Article

Techno-Economic Optimization of Hybrid Renewable Energy Systems (HRESs) and Feasibility Study on Replacing Diesel and Photovoltaic Systems with Hydrogen for Electrical and Small Deferrable Loads: Case Study of Cameroon

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Abstract

To reduce the amount of harmful gases produced by fossil fuels, more environmentally friendly and sustainable alternatives are being proposed around the world. As a result, technologies for manufacturing hydrogen fuel cells and producing green hydrogen are becoming more widespread, with an impact on energy production and environmental protection. In many countries around the world, and in Africa in particular, leaders, scientists, and populations are considering switching from fossil fuels to so-called green energies. Hydrogen is therefore an interesting alternative that deserves to be explored, especially since both rural and urban populations have shown an interest in using it in the near future, which would reduce pollution and the proliferation of greenhouse gases, thereby mitigating global warming. The aim of this paper is to determine the hybrid energy system best suited to addressing the energy problem in the study area, and then to make successive substitutions of different energy sources, starting with the most polluting, in order to assess the possibilities for transitioning the energy used in the area to green hydrogen. To this end, this study began with a technical and economic analysis which, based on climatic parameters, led to the proposal of a PV/DG-BATTERY system configuration, with a Net Present Cost (NPC) of USD 19,267 and an average Cost Of Energy (COE) of USD 0.4, and with a high proportion of CO₂ emissions compared with the PV/H₂GEN-BATT and H₂GEN systems. The results of replacing fossil fuel generators with hydrogen generators are beneficial in terms of environmental protection and lead to a reduction in energy-related expenses of around 2.1 times the cost of diesel and a reduction in mass of around 2.7 times the mass of diesel. The integration of H₂GEN, at high duty percentages, increases the Cost Of Energy, whether in a hybrid PV/H₂GEN system or an H₂GEN system. This shows the interest in the study country in using favorable duty proportions to make the use of hydrogen profitable.

Keywords: climatic conditions; hybrid systems; optimization; hydrogen



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1. Introduction

The development of the energy sector is one of the priorities for most governments around the world [1]. Major world powers are undertaking large-scale projects to build

power plants and transport routes to address their energy issues [2,3]. Underdeveloped and developing countries are no exception, especially since USD 3.4 trillion of investment would be needed to fund the infrastructure gap in Africa alone. It is extremely important to ensure that renewable energy projects are funded in an inclusive manner, i.e., by including any technology or fuel that is important for the future [4]. Power plants often occupy large areas and generally produce greenhouse gases, harmful gases, and pollution [4], which contribute to environmental problems such as global warming. With global warming increasing over the years, these countries will be forced to turn to so-called renewable energies [4].

The widespread use of hydrogen fuel cell technologies and the production of green hydrogen have an impact on energy production and environmental protection. In addition, the harmful gases produced by the use of fossil fuels make it necessary to push forward research into more environmentally friendly and sustainable alternatives. As a result, most countries around the world are considering replacing polluting energy sources with so-called green energies. Hydrogen is therefore an interesting alternative that is already being explored in developed countries, and exploration in Africa is imperative and would help reduce and prevent pollution and the proliferation of greenhouse gases, thereby mitigating global warming.

Geologically and geographically available renewable energy sources are relatively abundant and free, which is hardly the case for fossil fuels.

The use of solar and wind energy resources increased significantly between 1996 and 2013, rising from less than 50.0 GW in 1996 to just over 300.0 GW in 2013 [5].

Several scientific studies [4–24] have demonstrated the reliability and cost-effectiveness of using Hybrid Renewable Energy Systems (HRESs) and the benefits of considering hydrogen over fossil fuels to meet electricity needs (Tables 1–3). Table 1 shows that HRESs exist worldwide. Table 2 shows that in America, Europe, and Asia, numerous studies are being conducted on HRESs using hydrogen. Table 3 shows that studies are also underway in Africa on the use of HRESs using hydrogen.

Table 1. Literature review on PV/DG-BATT-HRES.

N°	Authors	Location	Grid Type
1	Lau et al., 2010 [6]	Malaysia	/
2	Adaramola et al., 2014 [7]	North of Nigeria	/
3	Jenkins et al., 2020 [8]	USA	/
4	Vakili-Nezhaad G et al., 2021 [9]	Desalination plant, Oman	Off Grid
5	Halabi, L. M. et al., (2017) [10]	Malaysia	/
6	Thirunavukkarasu and Sawle, 2021 [11]	India	/
7	Ayan and Turkay, 2023 [12]	Turkey	/
8	Li et al., 2022 [13]	China	Off Grid
9	Amoussou et al., 2024 [14]	Cameroon	/

The aim of this paper is to determine the hybrid energy system best suited to addressing the energy problem in the study area, and then to make successive substitutions of different energy sources, starting with the most polluting, in order to assess the possibilities for transitioning the energy used in the area to green hydrogen. The applications considered do not have access to the electricity grid, and battery storage is used in the assessment of the reliability of the proposed systems. Household electricity consumption is 7.7 kWh/day with a peak of 1.4 kW, and the deferrable load is 1.5 kWh/day with a peak of

0.2 kW (Table 4). The possibility of using green hydrogen as a clean fuel to replace diesel is being studied here in order to reduce and prevent atmospheric pollution and the proliferation of greenhouse gases with the aim of mitigating their effects on the environment (global warming).

Table 2. Literature review on HRESs with hydrogen.

N°	Authors	Location	Grid Type
1	Fazelpour et al., 2014a [5]	Teheran (Iran)	Off Grid
2	Turkay and Telli, (2011) [15]	Turkey	/
3	Moghaddas et al., 2019 [16]	Iran	/
4	Mansour et al., 2020 [17]	/	/
5	Yan et al., 2024 [18]	/	/
6	Lambert et al., 2021 [19]	Pakistan	On and Off Grid
7	Gharibi et al., 2025 [20]	/	/

Table 3. Literature review on HRESs using hydrogen in Africa.

N°	Authors	Location	Grid Type
1	Ringel et al., 2024 [21]	Egypt	/
2	Bahou, 2023 [22]	Moroco	
3	Amrani et al., 2023 [23]	Moroco	Off Grid
4	Okundamiya, 2021 [24]	Nigeria	
5	El Manaa et al., 2022 [25]	Tunisia	/
6	Gougui et al., 2020 [26]	Ghana	

Table 4. Power demand assessment of households in the study area.

Appliances	Number	Power (W)	Runtime	Energy Consumption (Wh)	Energy Consumption (kWh)
Freezer	1	170.0	10.0	1700.0	1.7
TV	1	56.0	12.0	672.0	0.7
Flat iron	1	30.0	1.0	30.0	0.1
Wi-Fi Box	1	5.0	3.0	15.0	0.1
Iron	1	1000.0	1.0	1000.0	1.0
	1	2200.0	1.0	2200.0	2.2
Computer	1	130.0	4.0	520.0	0.5
	1	65.0	4.0	260.0	0.3
Blender	1	350.0	0.1	28.0	0.1
	1	450.0	0.1	36.0	0.1
LED Bulb	10 × 15	150.0	5.0	750.0	0.8
	7 × 7	49.0	5.0	245.0	0.2
Telephone	1	33.0	3.0	99.0	0.1
	1	15.8	3.0	47.5	0.1
	1	33.0	3.0	99.0	0.1
	1	6.8	3.0	20.4	0.1
				Total	7.7

2. Materials and Methods

2.1. Climate Data

Located at coordinates 2° to 13° north latitude and 9° to 16° east longitude, Cameroon still faces electrification problems in both urban and rural areas, which could affect the country's sustainable development. Between 2014 and 2021, the rate of access to electricity in Cameroon increased by nearly 2%, from 58.4% to 60.2%, according to the results of the 5th Cameroon Household Survey, published on 24 April 2024, by the National Institute of Statistics (INS) [27]. The rate of access to electricity reaches 87.6% in urban areas of Cameroon, but is only 28.4% in rural areas [28]. This situation highlights the marked imbalance in electrification between cities and the countryside, making rural populations the most disadvantaged in this sector. The lack of energy services in rural areas and the ongoing energy crisis in urban and peri-urban areas, the resurgence of the harmful effects of climate change, the widespread shift to renewable energy, and Cameroon's strong potential for renewable energy mean that Cameroon should consider the effective deployment of renewable energy, despite the needs expressed and all the development challenges, the assessment of the political and legal framework with regard to the situation of renewable energies in Cameroon, and in comparison with the situation in other countries. In fact, Cameroon receives an average of $4.4 \text{ kWh/m}^2/\text{day}$ in terms of solar radiation, with a temperature of 21.6°C and an average wind speed of 3.0 m/s .

Given that the availability of energy potential depends on the geography of the area studied [5], Homer Pro software (3.16.2 version) was used to extract solar, wind, and temperature data from different areas of the country, and an interpolation was made in ArcGIS 10.8 software. Figure 1 shows the distribution of the three natural parameters that enable the configuration of renewable energy systems.

2.1.1. Solar Radiation

Solar radiation data for the various points considered are obtained using Homer Pro software. The solar radiation incident on the horizontal surface of the study area is $4.6 \text{ kWh/m}^2/\text{day}$. January and February are the months with the highest solar radiation, while August has the lowest. Figure 1a shows the variations in solar radiation in Cameroon. The maximum and minimum values are $5.7 \text{ kWh/m}^2/\text{day}$ and $3.1 \text{ kWh/m}^2/\text{day}$, respectively. High radiation levels are observed in the north, making this area suitable for photovoltaic energy exploitation. In addition, a decreasing variation in solar radiation can be observed from the north to south of the study area, with lower variations in the central part of the southern equatorial zone.

2.1.2. Wind Speed

The average annual wind speed for the study area is around 3.1 m/s . These values are measured and recorded at a height of 10 m. The highest average monthly wind speed occurs in August [29], and average wind speeds for the months of the year range from 1.7 m/s to 1.8 m/s . Figure 1b shows the different variations in wind speeds in the area. The maximum and minimum values are 5.8 m/s and 0.3 m/s , respectively. Wind speeds decrease from north to south, with minimum values in the central part of the south and southeast of the equatorial zone. The strongest winds are observed between 10° and $13^{\circ}05'$ north latitude in the northern part, making this the most suitable area for installing wind turbines.

2.1.3. Temperature

Temperature data for the study area were obtained using Homer Pro software, with the highest temperatures ranging from 24.1 (in March) to 24.0 (in February) degrees Celsius.

The lowest temperatures were observed in July (21.9 °C). Figure 1c shows the temperature variations in the area. The maximum and minimum values are 27.7 °C and 15.5 °C, respectively. The high solar radiation observed in the north may mean that this part could be classified as an area with high potential for photovoltaic energy [30]. The spatial variation in solar data and the spatial variation in temperature may be positively correlated with each other.

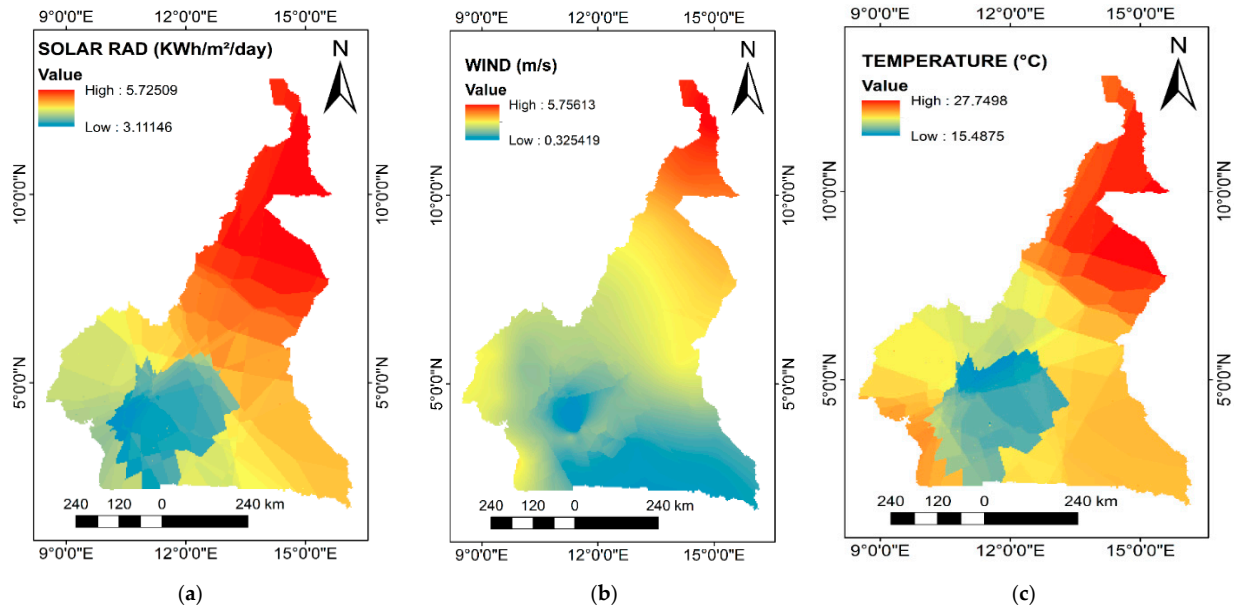


Figure 1. Distribution of resources: (a) solar radiation, (b) wind speed, (c) temperature.

2.2. Organization Chart

The organization chart below summarizes all of the methodological information used to carry out this work (Figure 2).

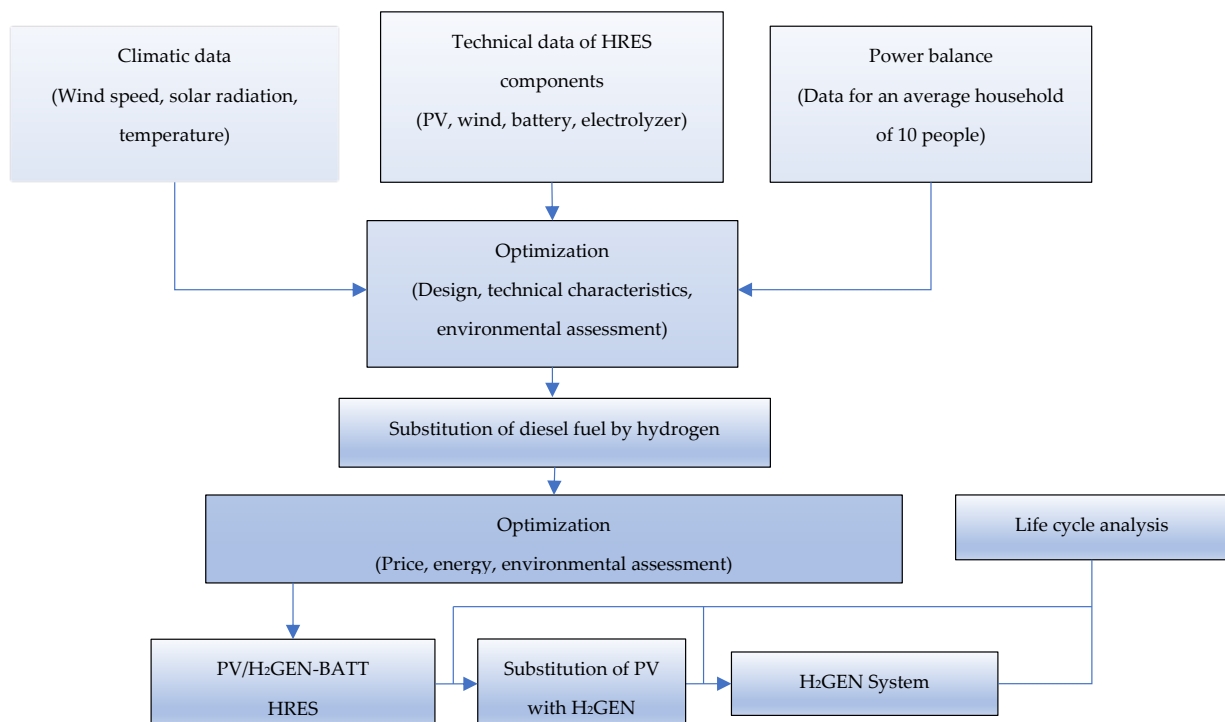


Figure 2. Illustrative organization chart of the method.

2.3. Homer Pro Software

Designed to simulate and optimize hybrid electrical systems, the Homer Pro software (3.16.2 version) is a versatile package. Based on the best possible match between supply and demand, this software package performs a large number of hourly simulations. It assesses the technical feasibility of a system to determine whether it can meet electrical and thermal loads while determining the total NPC of the system, expressed by Equation (1) [31].

$$NPC = \frac{C_{tot}}{CRF(i, T_p)} \quad (1)$$

where C_{tot} represents the total annual cost of the system (in USD/year), i is the annual interest rate (%), T_p is the project lifetime, and CRF is the capital recovery factor, expressed as Equation (2).

$$CRF = \frac{i(1+i)^n}{(1+i)^n - 1} \quad (2)$$

where n represents the number of years of the project. In the Homer Pro software, the payback cost is calculated as Equation (3).

$$SC = \frac{C_{rc} T_{rem}}{T_{com}} \quad (3)$$

where C_{rc} represents the replacement cost of the equipment (USD), T_{rem} represents the remaining life of the equipment (years), and T_{com} represents the life of the component (years). In addition, the discounted Cost Of Energy (COE) Equation (4) is calculated as follows:

$$COE = \frac{C_{tot}}{E_{tot}} \quad (4)$$

where E_{tot} is the total annual electricity consumption (kWh/year).

The environmental analysis was carried out using life cycle assessment. In this analysis, greenhouse gas emissions linked to the production of green hydrogen were considered negligible [32] and other emissions linked to the production of system components were taken into account.

3. Results

3.1. System Configuration

The study area was divided into four geographical regions: the Greater North (including the cities of Garoua, Maroua, and Ngaoundere), the Greater South (including the cities of Ebolowa, Douala, and Yaounde), the Greater East (including the city of Bertoua), and the Greater West (including the cities of Bafoussam, Buea, and Bamenda). Table 5 presents the data for these different cities.

The components of the system used as input data in the Homer Pro software are summarized in Table 6. These components and their characteristics are retained and used as input data for other regions of Cameroon to determine the profitability of the system based on climate data (sun, wind, and temperature).

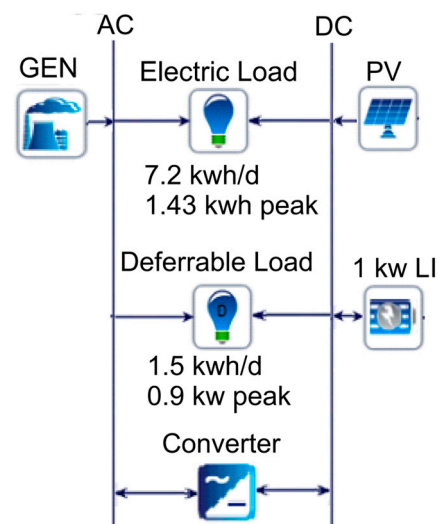
The system configuration will be based not only on climatic conditions to ensure the judicious and objective choice of components, but also on demand; the system is intended to be disconnected from the electricity grid in order to provide reliable and low-cost energy to households. Figure 3 shows the system configuration adapted to the geographical boundaries of the areas under consideration.

Table 5. Geographical location of the study areas.

		Location		Resources		
		Longitude E	Latitude N	Solar	Wind	Temperature
Greater South	Yaounde	3°51.7′	11°31.2′	4.67	2.07	22.97
	Ebolowa	2°55.6′	11°9.4′	4.64	1.76	22.83
	Douala	4°3.1′	9°46.1′	4.29	2.71	24.93
Greater West	Buea	4°9.4′	9°15.8′	4.29	3.12	25.10
	Bamenda	5°57.8′	10°9.5′	4.97	2.54	20.72
	Bafoussam	5°28.9′	10°25.6′	4.97	2.45	21.69
Greater East	Bertoua	4°34.8′	13°40.6′	5.02	2.60	23.53
Greater North	Ngaoundere	7°20.3′	13°34.0′	5.56	3.82	22.14
	Maroua	10°35.6′	14°19.3′	5.71	5.09	27.62
	Garoua	9°19.4′	13°23.6′	5.75	3.85	27.71

Table 6. Input data in Homer for the PV/DG-Batt HRES system.

Components	Model/Size	Lifetime	Capital Cost (USD/kW)	Replacement (USD/kW)	O&M	Quantity
Diesel Generator	Auto size Genset	15,000 h	1000	900	0.02 USD/hr	1
PV Modules	0, 1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, 10	20 years	3975	3378.7	0.00 USD/KW/year	-
Battery	Generic 1 KWh Li-ion (6 V, 167 h)	10 years	300	300	10.0 USD/year	0, 1, 2, 4, 6, 8, 10
Converter	Generic large, free conv	15 years	700	550	100.0 USD/year	0, 5, 10, 15, 20, 25

**Figure 3.** System configuration.

The input data in the Homer Pro software for the analyses of these systems are listed in Table 6. Note that these analyses performed by the software will provide the best-case scenario for each system. The monetary values in the analysis are in US dollars (USD) as of 30 January 2025; with a value of USD 1.3/l for diesel.

3.1.1. Technical and Economic Aspects

After analyzing the input data in the software, the results are presented according to two aspects that allow conclusions to be drawn about the importance of applying HRESs in the Central African sub-region, particularly in Cameroon.

Table 7 presents the results of the best scenarios (the results of optimization using Homer Pro software) for each region in terms of Net Present Cost (NPC), Cost Of Energy (COE), Operation and Maintenance Cost (O&M cost), and Renewable Fraction (RF) in the 10 regions of the country.

Table 7. Results of PV-DG-Batt based HRES.

Area	Town and System	Capital Cost (USD)	Replacement (USD)	O&M (USD)	Fuel (USD)	Salvage Cost (USD)	Total (USD)
Greater South	Yaounde PV/DG-BATT-Based HRES	10,500.0	2499.1	1877.9	5247.5	242.7	20,367.2
	Douala PV/DG-BATT-Based HRES	11,750.0	2470.3	1898.4	4840.6	272.6	21,232.0
	Ebolowa PV/DG-BATT-Based HRES	10,500.0	2508.1	1885.9	5334.7	232.2	20,461.0
Greater West	Buea PV/DG-BATT-Based HRES	11,750.0	2471.9	1898.4	4845.5	270.5	21,236.4
	Bafoussam PV/DG-BATT-Based HRES	10,500.0	2464.9	1817.7	4714.6	276.4	19,773.7
	Bamenda PV/DG-BATT-Based HRES	10,500.0	2464.4	1820.1	4731.6	277.5	19,793.8
Greater East	Bertoua PV/DG-BATT-Based HRES	12,262.5	5618.5	7978.6	9638.3	1531.9	37,029.9
Greater North	Ngaoundere PV/DG-BATT-Based HRES	10,500.0	2508.1	1885.9	5334.7	232.2	20,461.0
	Maroua PV/DG-BATT-Based HRES	10,500.0	2141.2	1655.1	3250.6	209.8	17,756.8
	Garoua PV/DG-BATT-Based HRES	10,500.0	2499.1	1877.9	5427.5	242.7	20,547.2

The overall architecture, regardless of the region, shows that the best configuration proposed is the PV–Diesel–Battery option with an average NPC of USD 19,267 and an average COE of USD 0.4/kWh.

It consists of a Generic flat Photovoltaic (PV) module (2.5 kW), Auto Genset (1.6 kW), Generic 1.0 kWh li-ion, a system converter (1.0 kW), and cyclic charging.

The Net Present Cost (NPC) values range from USD 17.3 to USD 20.6, with an average of USD 19.3. The COE values vary between USD 0.39 and USD 0.47/kWh, with an average of USD 0.4/kWh, while O&M costs vary between USD 528 and USD 734/year, with an average value of USD 648/year while the RF (Renewable Fraction) varies between 58.8% and 75.0% for an average value of 65.7%. These results show that, regardless of the location

within the country, the variations in all parameters (NPC, COE, and O&M) are small and the capital remains virtually the same for most cities.

Details of the NPC breakdown for different cities according to the system that was deemed the best scenario during the Homer analysis, namely PV–Diesel–Battery, are shown in Table 8.

Table 8. Breakdown of NPC for PV-Diesel Bat.

Area	Town and System	CAPITAL (USD)	Replacement (USD)	O&M (USD)	Fuel (USD)	Salvage Cost (USD)	Total (USD)
Greater South	Yaounde PV/DG-BATT-Based HRES	10,500.0	2499.1	1877.9	5247.5	106.1	20,230.5
	Douala PV/DG-BATT-Based HRES	11,750.0	2470.3	1898.4	4840.6	272.6	21,232.1
	Ebolowa PV/DG-BATT-Based HRES	10,500.0	2508.1	1885.9	5334.7	232.2	20,461.0
Greater West	Buea PV/DG-BATT-Based HRES	11,750.0	2471.9	1898.4	4845.5	270.5	21,236.4
	Bafoussam PV/DG-BATT-Based HRES	10,500.0	2464.9	1817.7	4714.6	276.4	19,773.7
	Bamenda PV/DG-BATT-Based HRES	10,500.0	2464.4	1820.1	4731.6	277.5	19,793.8
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	Maroua PV/DG-BATT-Based HRES	10,500.0	2141.2	1655.1	3250.6	209.8	17,756.8
	Garoua PV/DG-BATT-Based HRES	10,500.0	2499.1	1877.9	5427.5	242.7	20,547.2

Figures 4 and 5 provide more information on the distribution of system costs.

The high cost of the system for each city is due to the high cost of the battery and PV module compared with other components. The system's electricity production is mainly generated by the PV module, as shown in Figure 6. The diesel generator therefore acts as an alternative solution (preventive in case the PV does not meet the load) in electricity production, which means that the PV alone could meet the demand in each city according to the trend shown in Figure 6.

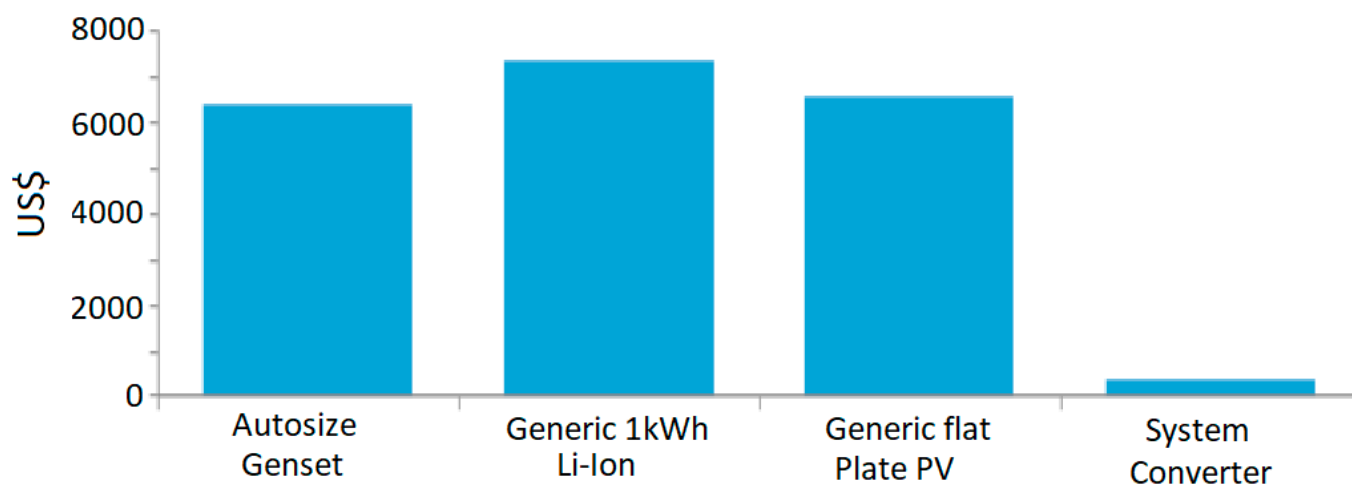


Figure 4. Cost summary of Net Present Cost by components (Homer Pro software).

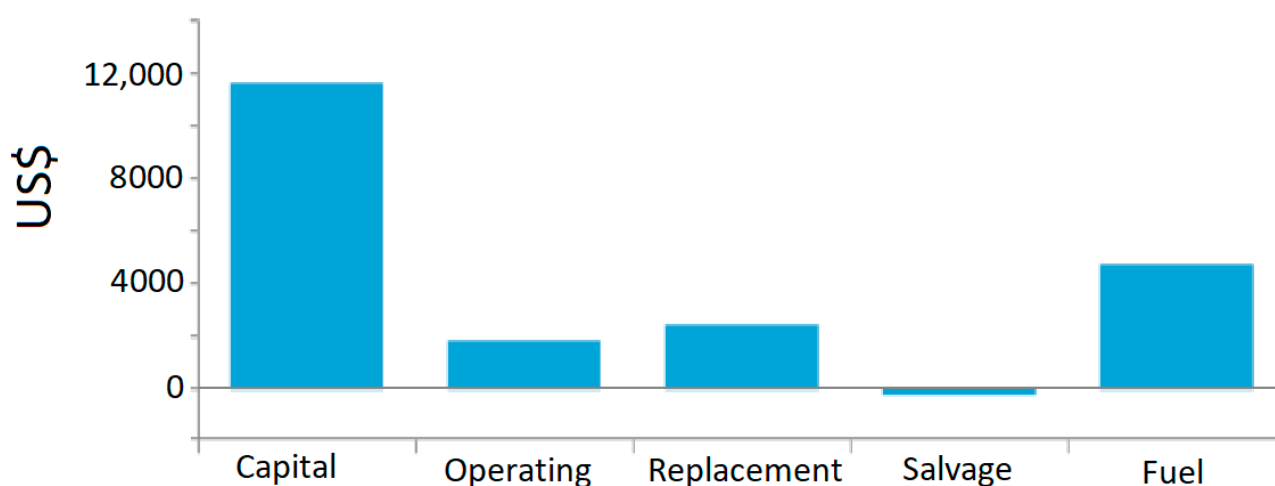


Figure 5. Cost summary of Net Present Cost by cost type (Homer Pro software).

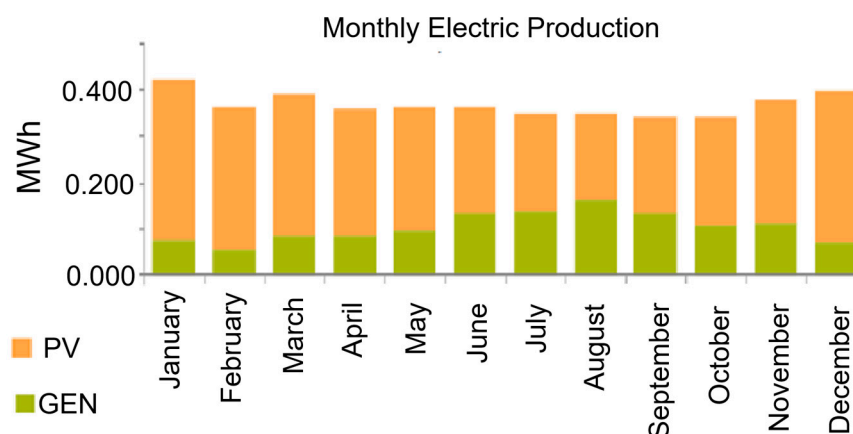


Figure 6. Electrical production via the PV and the DG (Homer Pro software).

3.1.2. Environmental Aspect

The environmental aspect here is based on the assessment of the proportions of waste from the use of the diesel generator, in particular CO₂ and other harmful gases, as shown in Table 8. Although most of the electricity is generated by the photovoltaic module, the impact of the generator's emissions on the environment should not be overlooked. Comparing the masses of pollutant emissions, we see an average emission of 1017.2 kg/year

for CO₂, 6.5 kg/year for CO, 0.2 kg/year for UHC, 0.1 kg/year for PM, 2.4 kg/year of SO₂, and 6.0 kg/year of NO_x for all ten regions in the area.

Table 9 illustrates the total amount of pollutants considered per year, and we can see that the proportion of CO₂ from the generator's fossil fuel (diesel) is higher than the others (CO, UHC, PM, SO₂, and NO_x), a finding also noted by [33], due to the fact that carbon is the main component of fuel [34].

Table 9. Environmental emissions.

Environmental Emissions							
Area	Towns	CO ₂ (Kg/Year)	CO (Kg/Year)	UHC (Kg/Year)	PM (Kg/Year)	SO ₂ (Kg/Year)	NO _x (Kg/Year)
Greater South	Yaounde	1063.0	6.7	0.29	0.04	2.6	6.3
	Douala	980.0	6.1	0.27	0.03	2.4	5.8
	Ebolowa	1080.0	6.8	0.29	0.04	2.6	6.4
Greater West	Buea	845.0	5.3	0.23	0.03	2.1	5.0
	Bamenda	958.0	6.0	0.26	0.03	2.3	5.7
	Baffoussam	955.0	6.0	0.26	0.03	2.3	5.6
Greater East	Bertoua	1490.0	9.3	0.41	0.05	3.6	8.8
Greater North	Ngaoundere	1080.0	6.8	0.29	0.04	2.6	6.4
	Maroua	658.0	4.1	0.18	0.02	1.6	3.9
	Garoua	1063.0	6.7	0.29	0.04	2.6	6.3

Cameroon, like other countries in the world and in sub-Saharan Africa, is moving towards the energy transition, and data collected on the possibility of using cleaner fuels such as green hydrogen, for example, would be a welcome, environmentally responsible alternative in the fight against global warming and its effects on the environment.

3.2. Replacing Different Energy Sources with Hydrogen Generators

Hybrid PV/DG-BATT HRESs are systems that produce significant amounts of greenhouse gases. Environmental protection objectives aim for near-zero greenhouse gas production. The most suitable energy sources for meeting these criteria are renewable energy sources, in general, and hydrogen generators in particular.

Technologies for producing green hydrogen and constructing hydrogen-powered electric generators have improved considerably, and the cost of producing electricity in this way has fallen significantly over the years, to the point where hydrogen generators are now affordable for people in both developed and developing countries [35]. In addition, 1.0 kg of hydrogen produces 2.5 to 3.0 times more energy than conventional fossil fuels [35]. In the first part of this work, it was demonstrated that the PV/DG-BATT HRES hybrid system is the most suitable among the energy sources already in use in the study area. The aim of this work is to propose a system that is both economical and environmentally friendly. To this end, two substitutions of the energy sources used in the PV/DG-BATT HRES were made to obtain a PV/H₂GEN-BATT system and then an H₂GEN system (Table 10). Technical and economic analyses were carried out for these two systems, including an assessment of the customs clearance price of equipment and hydrogen in this country. Customs duties were considered at 0%, 50%, and 100%.

Table 10. Input data.

Components	Model/Size	Lifetime	Capital Cost (USD)	Replacement	O&M (USD)	Quantity
H2 Generator	2 kw	25,000 h	1500.0 USD/kw [36]	1200.0 USD/kW	0.02	1
PV modules	0, 1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, 10	20 years	3975.0 USD/kW	3378.7 USD/kW	0.00	-
Battery	Generic 1 KWh Li-ion (6 V, 167 h)	10 years	300.0 USD/	300.0 USD/kW	10.00	0, 1, 2, 4, 6, 8, 10
Converter	Generic large, free conv	15 years	700.0 USD/	550.0 USD/kW	100.00	0, 5, 10, 15, 20, 25
Hydrogen tank	Generic 1, 2, 4, 6 kg	25 years	36.6 USD/ /kg (H2)	18.3 USD/kg (H2)	10.00	-
Electrolyzer	10, 15, 20 KW	15 years	2000.0 USD	1700.0 USD/kW	10.00	-

3.2.1. Replacing the Diesel Generator with a Hydrogen Generator

Replacing the GEN with the H₂GEN in the PV/DG-BATT hybrid system resulted in the PV/H₂GEN-BATT system (Figure 7a). Across the ten data collection sites, this system has an average NPC of 12,596.9 USD, 16,918.8 USD, and 21,233.4 USD (Table 11) and an average COE of 0.4 USD, 0.2 USD, 0.2 USD, and 0.3 USD, with customs duties considered at 0%, 50%, and 100% of the equipment price, respectively (Table 12). The fuel consumption of hydrogen generators was assessed in each city in the study area, and the average annual hydrogen consumption was 79.4 Kg/years for customs duties considered at 0%, 50%, and 100% of the equipment price (Table 13).

Table 11. NPC in USD for customs duties of 0%, 50% and 100%.

	PV/H ₂ GEN-BATT 0%	PV/H ₂ GEN-BATT 50%	PV/H ₂ GEN-BATT 100%	H ₂ GEN 0%	H ₂ GEN 50%	H ₂ GEN 100%
Garoua	12,683.4	17,037.6	21,391.8	14,661.2	21,991.9	29,322.5
Maroua	11,558.5	15,350.3	19,142.1	15,011.2	2,2516.8	30,022.4
Ngaoundere	12,735.4	17,115.6	21,495.8	14,657.9	21,985.4	29,313.8
Yaounde	12,683.4	17,037.6	21,391.8	14,661.2	21,991.9	29,322.5
Douala	12,457.5	16,698.7	20,940.0	15,304.6	22,956.9	30,609.3
Ebolowa	12,735.4	17,115.6	21,495.8	14,657.9	21,985.4	29,313.8
Bertoua	13,943.3	18,927.4	23,911.6	14,783.0	22,174.5	29,566.0
Buea	12,386.4	16,700.6	20,942.5	15,309.0	2,2963.5	30,618.0
Baffoussam	12,386.4	16,592.1	20,797.8	14,772.1	22,158.2	29,544.2
Bamenda	12,399.9	16,612.4	20,824.8	14,787.3	22,181.0	29,544.2
Average	12,596.9	16,918.8	21,233.4	14,860.5	22,178.1	29,717.7
Minimum	11,558.5	15,350.3	19,142.1	14,657.9	21,985.4	29,313.8
Maximum	13,943.3	18,927.4	23,911.6	15,309.0	22,956.9	30,618.0

Table 12. COE in USD for customs duties of 0%, 50% and 100%.

	PV/H ₂ GEN-BATT 0%	PV/H ₂ GEN-BATT 50%	PV/H ₂ GEN-BATT 100%	H ₂ GEN 0%	H ₂ GEN 50%	H ₂ GEN 100%
Garoua	0.2	0.2	0.3	0.2	0.3	0.5
Maroua	0.1	0.2	0.2	0.2	0.3	0.4
Ngaoundere	0.2	0.2	0.3	0.2	0.3	0.5
Yaounde	0.2	0.2	0.3	0.2	0.3	0.4
Douala	0.1	0.2	0.3	0.2	0.3	0.4
Ebolowa	0.2	0.2	0.3	0.2	0.3	0.4
Bertoua	0.2	0.3	0.4	0.2	0.3	0.5
Buéa	0.1	0.2	0.3	0.2	0.3	0.4
Baffoussam	0.1	0.2	0.3	0.2	0.3	0.5
Bamenda	0.1	0.2	0.3	0.2	0.3	0.5
Average	0.2	0.2	0.3	0.2	0.3	0.5
Minimum	0.1	0.2	0.2	0.2	0.3	0.4
Maximum	0.2	0.3	0.4	0.2	0.3	0.5

Table 13. Hydrogen (H₂) consumption in kg/year for customs duties of 0%, 50% and 100%.

	PV/H ₂ GEN-BATT 0%	PV/H ₂ GEN-BATT 50%	PV/H ₂ GEN-BATT 100%	H ₂ GEN 0%	H ₂ GEN 50%	H ₂ GEN 100%
Garoua	81.7	81.7	81.7	246.0	246.0	246.0
Maroua	50.6	50.6	50.6	255.6	255.6	255.6
Ngaoundere	83.1	83.1	83.1	245.8	245.8	245.8
Yaounde	81.7	81.6	81.7	246.0	246.0	246.0
Douala	75.4	75.4	75.4	263.7	263.7	263.7
Ebolowa	83.1	83.1	83.1	245.8	245.8	245.8
Bertoua	116.4	116.4	116.4	249.3	249.3	249.3
Buéa	75.4	75.4	75.4	263.8	263.8	263.8
Baffoussam	73.4	73.5	73.5	249.1	249.0	249.0
Bamenda	73.8	73.8	73.8	249.5	249.4	249.4
Average	79.4	79.4	79.5	251.5	251.5	251.5
Minimum	50.6	50.6	50.6	245.8	245.8	245.8
Maximum	116.4	116.4	116.4	263.8	263.8	263.8

3.2.2. Replacement of Diesel Generators and Photovoltaics with H₂GEN

Replacing PV with H₂GEN in the PV/H₂GEN-BATT hybrid system led to the H₂GEN system (Figure 7b). At the ten data collection sites, this system has an average NPC of 14,860.5 USD, 22,178.1 USD, and 29,717.7 USD (Table 11) and an average COE of 0.2 USD, 0.3 USD, and 0.4 USD for customs duties considered at 0%, 50%, and 100% of the equipment price, respectively (Table 12).

The fuel consumption of hydrogen generators was assessed in each city in the study area, and the average annual hydrogen consumption was 251.5, 251.5, and 251.5 kg/year for customs duties considered at 0%, 50%, and 100% of the equipment price, respectively (Table 13).

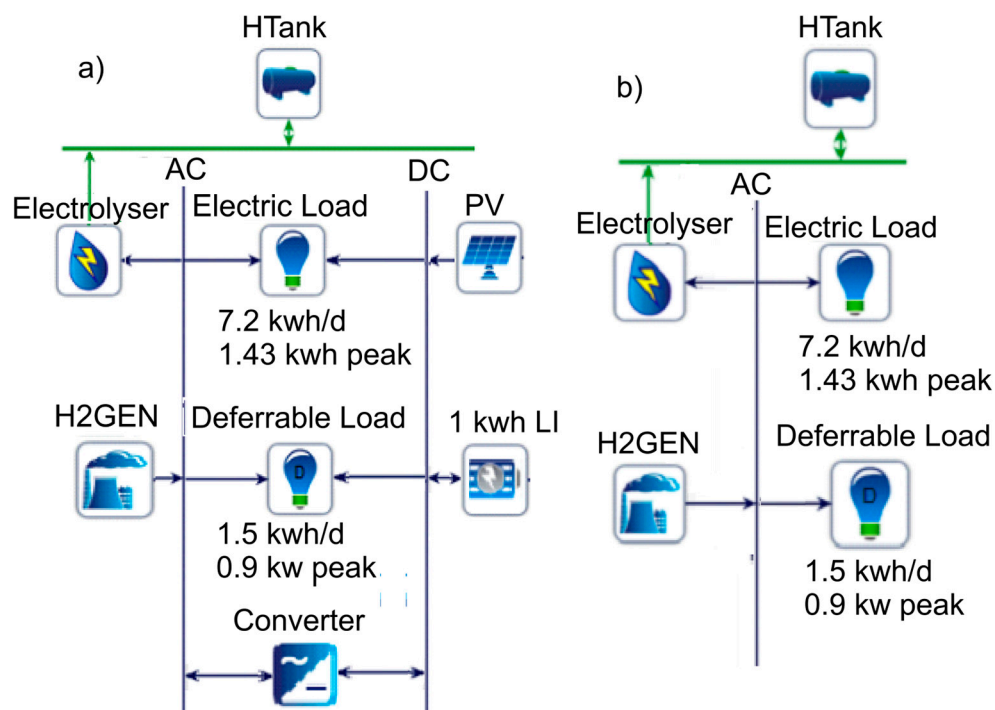


Figure 7. Systems obtained after substitution of GEN then PV, (a) PV/H₂GEN-BATT, (b) H₂GEN.

4. Discussion

The aim of this paper was to determine the hybrid energy system best suited to addressing the energy problem in the study area, and then to make successive substitutions of different energy sources, starting with the most polluting, in order to assess the possibilities for transitioning the energy used in the area to green hydrogen.

4.1. Best HRES (PV-DG-Batt HRES) Configuration for the Study Area

The optimization shows that the PV/DG-Batt system is currently the most suitable energy solution for the area [4–11]. It is estimated to have an average Net Present Cost (NPC) of USD 19,267 and an average Cost Of Energy (COE) of USD 0.4/kWh [5]. These values show little variation from one region to another. The photovoltaic module is the main source of electricity production (68.3% on average), while the diesel generator plays a complementary role (31.6% on average) in electricity production for the household. This suggests that PV alone could cover most of the energy demand, thereby reinforcing the renewable nature of the solution [6]. Despite the backup role of diesel, polluting emissions are inevitable [11]. Maximum values of 1080 kg/year of CO₂, 3.4 kg/year of CO, 0.3 kg/year of UHC, 0.04 kg/year of PM, 2.6 kg/year of SO₂, and 8.8 kg/year of NO_x were obtained for the ten regions in the study area. Replacing diesel with hydrogen is attractive, especially from an environmental perspective, as it significantly reduces the carbon footprint and preserves sustainable living conditions, thereby strengthening environmental protection worldwide, since hydrogen combustion produces only water [18].

4.2. Successive Substitutions of Different Energy Sources by Hydrogen

The most cost-effective system with the most favorable operating and maintenance (O&M) costs is the PV/H₂GEN system, while the most expensive system to maintain is the H₂GEN system with customs duties at 100% of the purchase price of the equipment. The PVH₂GEN system has lower O&M costs than the PV/DG-BATT system for customs duties ranging from 0% to 100% of the price of equipment and fuel (hydrogen). The H₂GEN system remains more favorable than the PV/DG-BATT system when customs duties are 0%. This is no longer the case above 50% (Figure 8a). Similarly, the system with the most favorable COE is PV/H₂GEN with 0% customs duties, followed by the H₂GEN system with 0% customs duties. The PV/H₂GEN and H₂GEN systems all have lower COEs than the PV/DG-BATT system, except for H₂GEN with 100% customs duty (Figure 8b).

The PV/DG-BATT hybrid system appears to be a very costly solution when looking at the NPC values (Figure 8c). We can see that the PV/H₂GEN and H₂GEN systems with 0% customs duties remain the most favorable.

The annual mass consumption of hydrogen is lower than that of diesel (Figure 8d, Table 12) [21]. Considering a purchase price of hydrogen at USD 2.4/kg, after tracing the trend curve, based on data aligned with China's [37,38], we see that for the same energy demand, the costs generated by hydrogen are lower than those imposed by diesel, even when customs duties are set at 100% (Figure 8e, Table 12). The results of replacing fossil fuel generators with hydrogen generators are beneficial in terms of environmental protection and lead to a reduction in energy costs of around 2.1 times the cost of diesel and a reduction in mass of around 2.7 times the mass of diesel. The integration of hydrogen, particularly at high duty cycles (100%), significantly increases the Cost Of Energy, whether in a hybrid PV/H₂GEN-Batt system or a hydrogen-only system. This shows the benefit for this country of using favorable duty cycles to make the use of hydrogen cost-effective [22].

4.3. Environmental Aspect

The life cycle analysis [39,40] enabled the assessment of greenhouse gas emissions in CO₂ equivalents in PV/DG-BATT, PV/H₂GEN-BATT, and H₂GEN systems in the 10 regions. The results show that the PV/DG-BATT system (photovoltaic, conventional thermal generator, and batteries) emits a significantly high amount of CO₂ (14,585.3 kg), highlighting the dominant impact of the fossil fuel-powered thermal generator [38]. In contrast, the PV/H₂GEN-BATT system (photovoltaic, hydrogen generator, and batteries) has a drastically reduced carbon footprint (2316.6 kg), demonstrating the environmental advantage of replacing a conventional generator with hydrogen technology, especially if the hydrogen is produced in a green manner [38]. The H₂GEN system alone (hydrogen generator) has a slightly higher footprint than the PV/H₂GEN-BATT hybrid system (2492.1 kg), suggesting that integrating photovoltaics and battery storage can further optimize the overall carbon footprint, potentially by reducing the hydrogen generator's dependence on hydrogen or by producing hydrogen more cleanly. In summary, conventional generators are the main contributors to CO₂ emissions, while the adoption of systems integrating photovoltaics and green hydrogen represents a major pathway for the significant decarbonization of energy production (Figure 9 and Table 14) [38].



Figure 8. Results of the technical and economic analyses of the systems (a) O&M, (b) COE, (c) NPC, (d) Consumption overview, (e) Price overview.

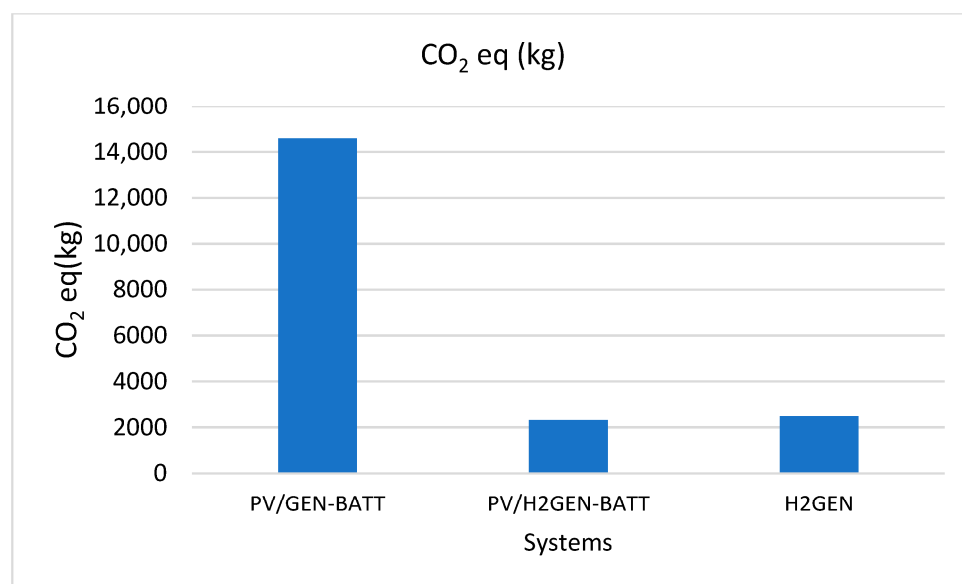


Figure 9. Results of environmental analyses of the systems.

Table 14. CO₂ equivalents of the systems.

Systems	CO ₂ EQ (kg/Year)
PV/GEN-BATT	14,585.4
PV/H ₂ GEN-BATT	2316.6
H ₂ GEN	2492.1

5. Conclusions

The aim of this paper was to determine the hybrid energy system best suited to addressing the energy problem in the study area, and then to make successive substitutions of different energy sources, starting with the most polluting, in order to assess the possibilities for transitioning the energy used in the area to green hydrogen. This study identified the PV/DG-BATtery hybrid system as the most suitable energy solution for the study area, with a Net Present Cost (NPC) of USD 19,267 and an average Cost Of Energy (COE) of USD 0.4. However, this initial configuration has a high proportion of CO₂ emissions. The substitution analysis clearly demonstrated that replacing fossil fuel (diesel) generators with hydrogen (fuel cell) generators offers significant advantages. Not only does this contribute to better environmental protection, but it also generates a reduction in energy expenditure of around 2.1 times less than the annual cost of diesel and a reduction in fuel mass of around 2.7 times less than that of diesel. Despite the environmental and economic benefits, replacing PV and diesel generators is more advantageous than the P-Batt system when the hydrogen tax is low. The integration of H₂GEN technology, especially at high customs duty rates, significantly increases the Cost Of Energy, whether in a hybrid PV/H₂GEN system or a pure H₂GEN system. It is therefore crucial that countries consider favorable customs duties in order to fully capitalize on the use of hydrogen and accelerate this essential energy transition for a more sustainable future.

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